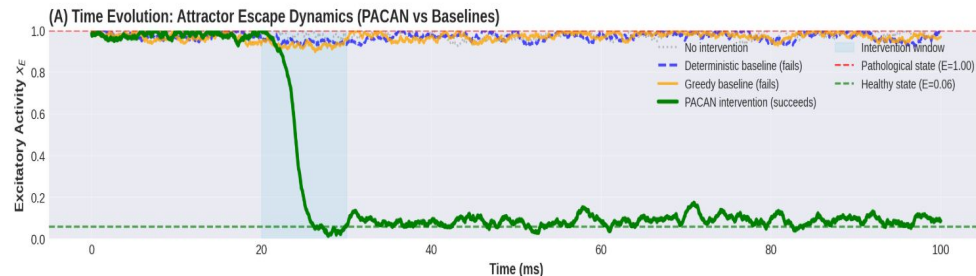


PACAN: Probabilistic Attractor Control for Aberrant Neural States

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W_C Network Ideal
Equations
(Dynamics for E and I)

Add Stochastic Noise
(Real Simulation)

Decide intervention Mask
(w_{ee}, w_{ii}, u_i)

Monte Carlo Simulation to see which attractor does the network settle to

- Assuming a WC Network, the Seizure-like activity* is modeled as a stable high-excitability E/I attractor that has noise; the goal is to shift neural population dynamics toward a lower-excitability healthy basin with minimum intervention.
- PACAN uses Boltzmann optimization technique to samples sparse intervention masks**, tests them through noisy Wilson–Cowan rollouts, and selects interventions that maximize P_{succ} , while minimizing energy and stimulated nodes. Tested $N = 100$ (200 nodes)
- The rollout is spread into 3 phases: pre_intervention, intervention, settlement

* Heitmann & Breakspear

** Ezaki et al., 2018

Results and What is next

Advantages Over Deterministic Policies(Greedy)	Neuroscience Interpretation
2x better Pareto hypervolume (Success, Nodes, Energy)	Finding sparse stimulation sites with minimum interventions
Pathological basin variance increases with large size (nodes)	The seizure-like state is not just a tiny corner. Dominant intervention is harder because the system naturally wants to return to the pathological state.
Reshape (GABA) > Push (Direct Current)	Interventions that alter E/I landscape geometry are 2.5X more robust than interventions that only add external drive.

1. Applying on CHB-MIT Dataset. But how??
2. What about multiple basins?

